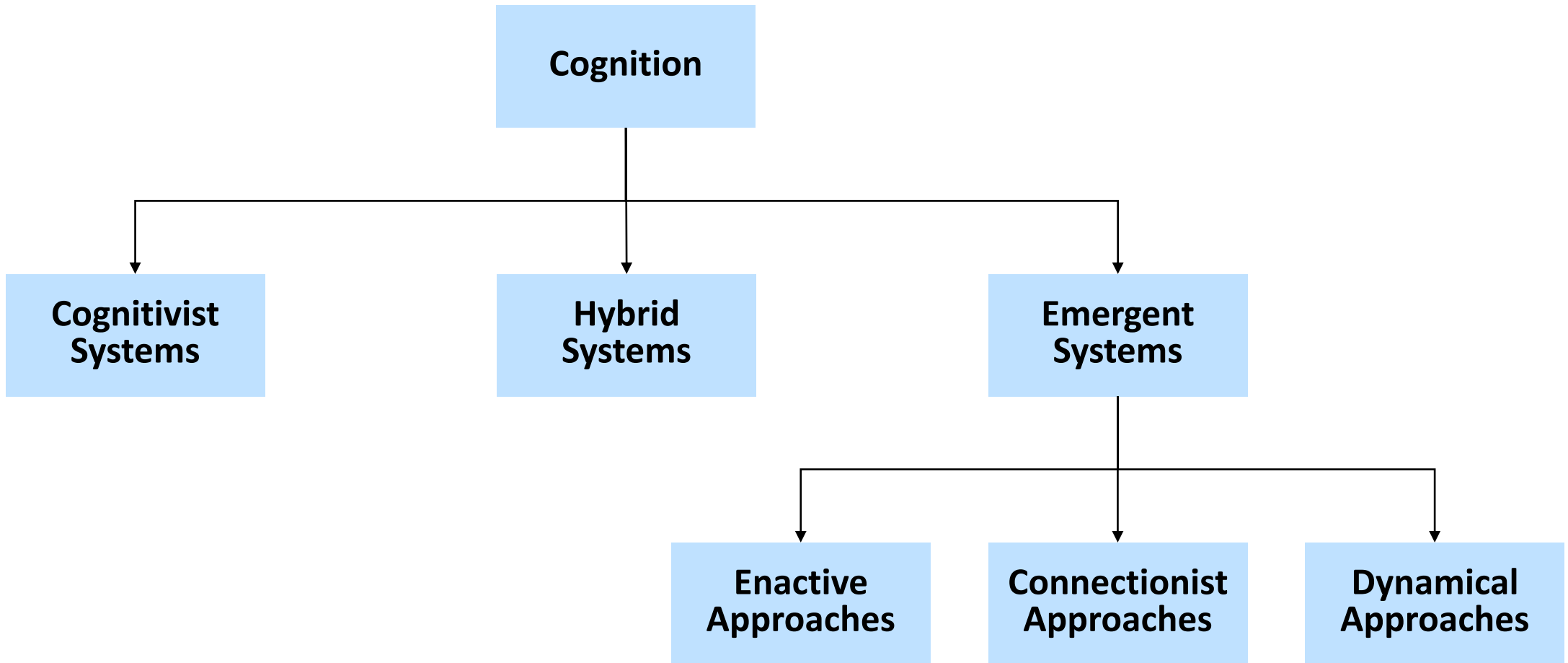


Modeling Cognitive Processes – Tools and Methods

Paradigms in Cognitive Science



From Vernon (2014)

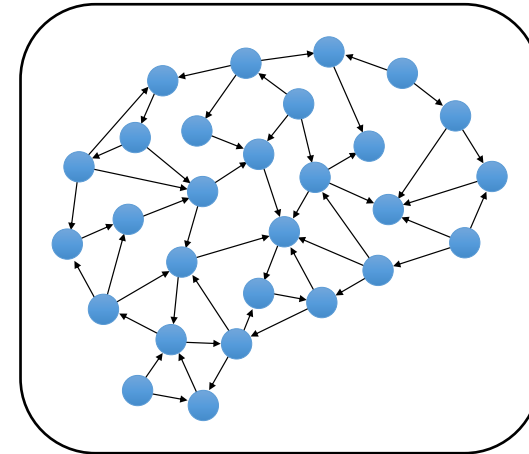
Paradigms of Cognitive Modeling

Cognitivist Systems

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Cognitivist cognitive modelling is based on the hypothesis that cognition is a form of **computation**. Cognitive functions are modelled as **working computer programs**.

Emergent Systems



In emergent cognitive models, cognition is a **continuous self-organizing process** that is driven by the interaction between the agent and its environment.

Cognitivism and AI

- Cognitivism started as a sister discipline of **Artificial Intelligence** (AI) during a summer school held at Dartmouth College in 1956
- The central hypothesis of cognitivism is that cognition is achieved by operations on **symbols** that have a one-to-one **correspondence** to real-world **actions** and **objects**
- External information from the environment is gathered and filtered by **perceptual processes**
- Over time, the focus of Artificial Intelligence shifted to **purely computational** techniques
- Artificial Intelligence with focus on cognition is today called **Artificial General Intelligence**

A Proposal for the

DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

June 17 - Aug. 16

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

The following are some aspects of the artificial intelligence problem:

1) Automatic Computers

If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and memory capacities of present computers may be insufficient to simulate many of the higher functions of the human brain, but the major obstacle is not lack of machine capacity, but our inability to write programs taking full advantage of what we have.

2) How Can a Computer be Programmed to Use a Language

It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning

Philosophy of Mind

Cognitivism postulates that all cognitive processes can be explained mechanistically through **some form of computation**. The philosophy of mind challenges this simplifying view by addressing – among many others – the following questions:

- What is the **basic nature of mental phenomena** such as thought, feeling, perception, consciousness and sensory experience?
- Can these phenomena be described solely by **chemical/physical processes** or can they **exist independently of them** (e.g. inside a computer)?
- How to explain **subjective conscious experiences (qualia)**?

What is the relation between the mind (mental world) and the body (physical world)?

Mind-Body Problem

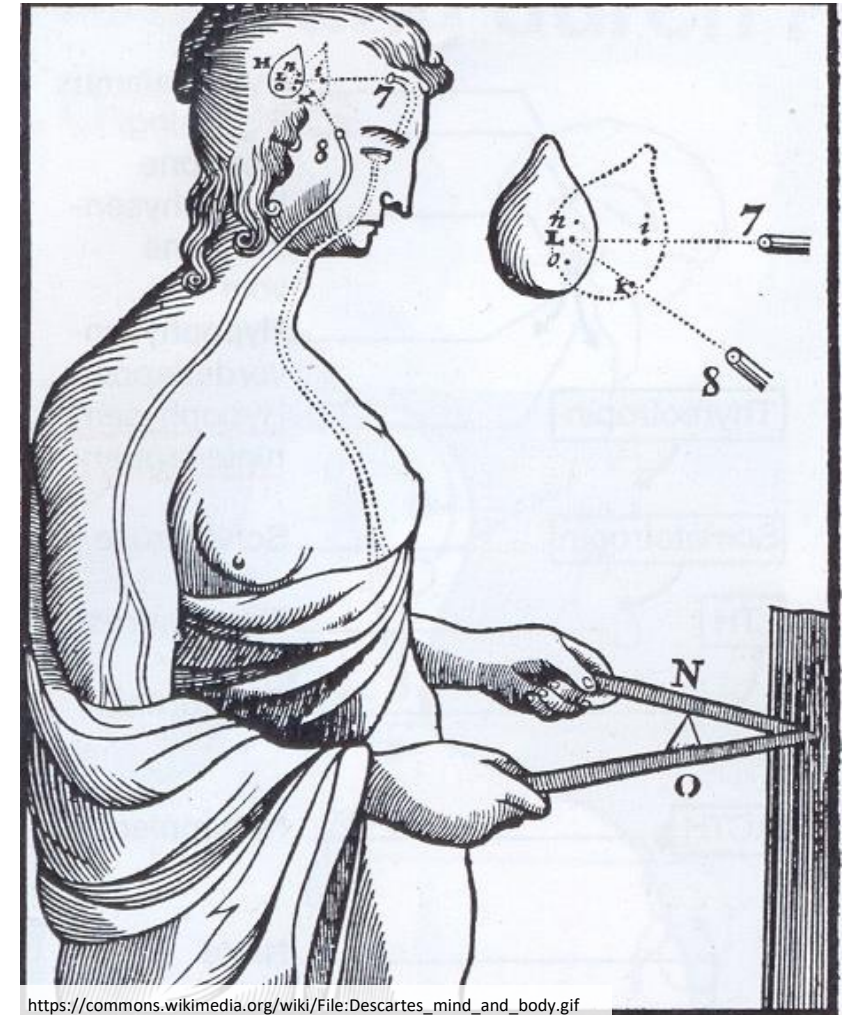
The mind-body problem addresses the question of how the **mental world** is related to the **physical world**. In general, there are two schools of thought:

Substance Dualism (also Cartesian Dualism)

- René Descartes postulated that mind and body are two kinds of different **substances**; their separation makes the **soul immortal** and enables **free will**
- Descartes falsely assumed that the pineal gland (Zirbeldrüse) interfaces the mind with the physical world

Monism (also Physicalism)

- There is **only one substance**, which means that **mental states are physical states**
- This would imply that individuals that **share a mental property also share a corresponding physical property**; nonreductive physicalism therefore postulates that mental properties are not identical physical ones even though there is only one substance



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Philosophy of mind

From Wikipedia, the free encyclopedia

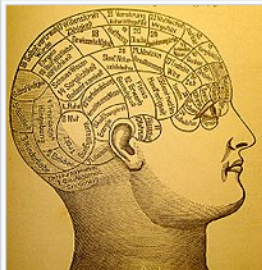
Philosophy of mind is a branch of [philosophy](#) that studies the [ontology](#) and nature of the [mind](#) and its relationship with the body. The [mind–body problem](#) is a paradigmatic issue in philosophy of mind, although a number of other issues are addressed, such as the [hard problem of consciousness](#) and the nature of particular mental states.^{[2][3][4]} Aspects of the mind that are studied include [mental events](#), [mental functions](#), [mental properties](#), [consciousness](#), the ontology of the mind, the nature of thought, and the relationship of the mind to the body.

[Dualism](#) and [monism](#) are the two central [schools of thought](#) on the mind–body problem, although nuanced views have arisen that do not fit one or the other category neatly.

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Philosophie des Geistes

Die **Philosophie des Geistes** ist ein Teilgebiet der [Philosophie](#) und beschäftigt sich mit der Natur [geistiger oder mentaler](#)^[1] Zustände, ihren [Wirkungen und Ursachen](#). Zentral ist dabei die Frage nach dem Verhältnis von geistigen und körperlichen Zuständen. Neben diesen [ontologischen](#) Fragen befasst sich die Philosophie des Geistes auch mit den [erkenntnistheoretischen](#) Fragen nach der [Erkennbarkeit](#) des Geistes. Die Philosophie der *Bewegung des Geistes durch die Geschichte* (wie sie beispielsweise in *Hegels Phänomenologie des Geistes* einen besonderen Höhepunkt fand) wird thematisch davon getrennt. Geist wird in der Philosophie des Geistes als *mind* verstanden und nicht als *Weitgeist*.

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- Das Leib-Seele-Problem
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Mind–body problem

From Wikipedia, the free encyclopedia

The **mind–body problem** is a debate concerning the relationship between thought and [consciousness](#) in the human mind, and the brain as part of the physical body. It is distinct from the question of how mind and body function chemically and physiologically, as that question presupposes an interactionist account of mind–body relations.^[1] This question arises when mind and body are considered as distinct, based on the premise that the mind and the body are fundamentally different in nature.^[1]

The problem was addressed by [René Descartes](#) in the 17th century, resulting in [Cartesian dualism](#), and by pre-Aristotelian philosophers,^{[2][3]} in [Avicennian philosophy](#),^[4] and in earlier Asian traditions. A variety of approaches have been proposed. Most are either [dualist](#) or [monist](#). Dualism maintains a rigid distinction between the realms of mind and matter. [Monism](#) maintains that there is only one unifying reality, substance or essence, in terms of which everything can be explained.

Each of these categories contains numerous variants. The two main forms of

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René Descartes' illustration of [mind/body dualism](#). Descartes believed inputs were passed on by the sensory organs to the [epiphysis](#) in the brain and from there to the

Mind-Body Problem

The *mind-body problem* is the problem of explaining how our mental states, events, and processes are related to the physical states, events, and processes in our bodies. A ques-

The Computational Theory of Mind

The computational theory of mind (CTM) is the theoretical basis of cognitivism:

“The **computational theory of mind** (CTM) holds that the mind is a **digital computer**: a **discrete-state** device that stores **symbolic representations** and manipulates them according to **syntactic rules**; that thoughts are mental representations – more specifically, symbolic representations in a **language of thought**; and that mental processes are causal sequences driven by the syntactic, but **not by the semantic**, properties of symbols. [...]”

MIT Encyclopedia of the Cognitive Sciences

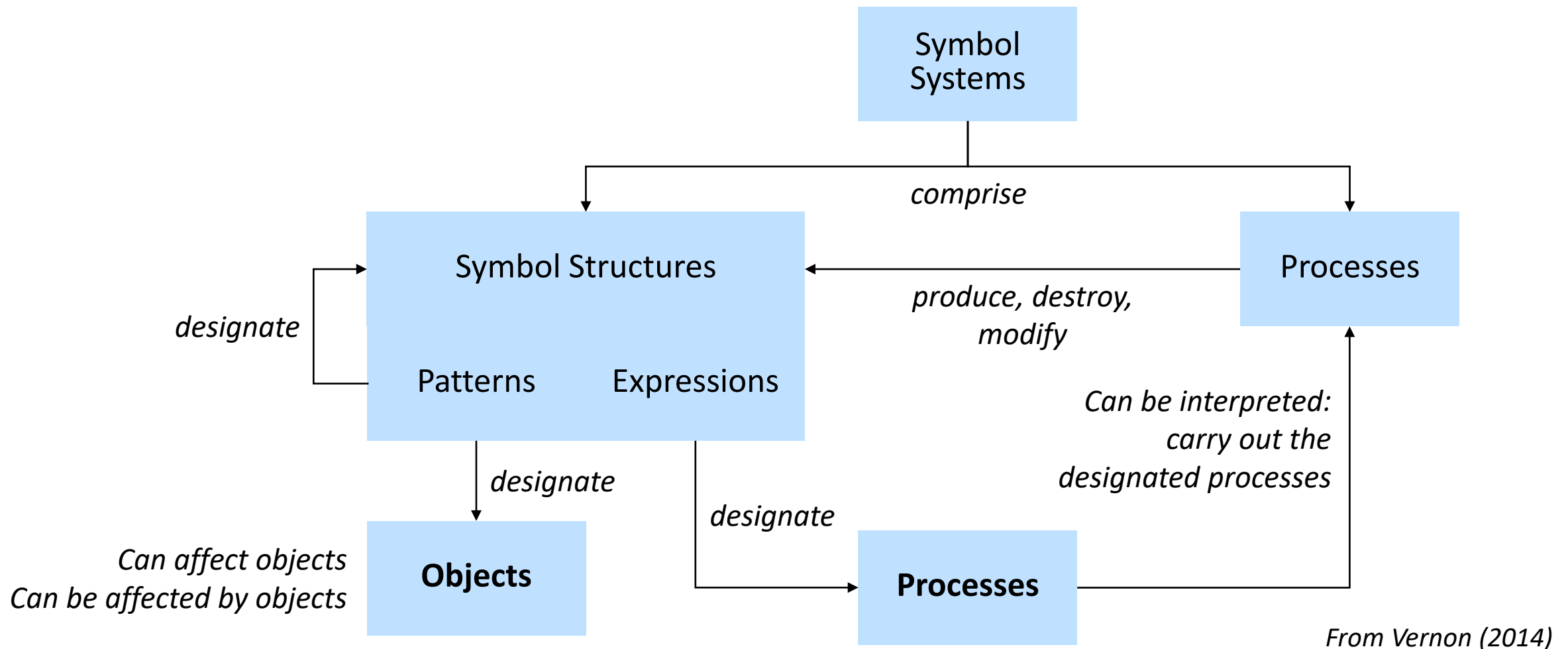
Computers are viewed as a proof that purely **syntactic operations** can express the causality in the **semantics** of cognitive processes. Cognition can therefore be implemented in a **purely physical system**.

Physical Symbol Systems

The concept of **Physical Symbol Systems** (PSS) by Newell and Simon lays the theoretical foundations for the emergence of general intelligence from symbolic operations:

- A PSS is comprised of **physical patterns** called **symbols** which can form **expressions** that are created, modified, copied, and destroyed by **processes**
- An expression **designates** an object iff the system can **affect it** or if it is **affected by it**
- The system can **interpret** an expression iff it designates a process that can be carried out by the system
- Both symbols and processes are **defined recursively**: Symbols can designate symbols and processes can produce other processes (→ **Development**)
- Problem solving is done by **searching**

Physical Symbol Systems



Two Hypotheses on Physical Symbol Systems

Physical Symbol System Hypothesis, Newell and Simon (1975)

A physical symbol system has the **necessary** and **sufficient** means for **general intelligent** action.

→ Every cognitive system is a physical symbol system

Heuristic Search Hypothesis, Newell and Simon (1975)

The solutions to problems are represented as symbol structures. A physical-symbol system exercises its intelligence in **problem-solving by search**, that is, by **generating** and progressively modifying symbol structures until it produces a **solution structure**.

→ Cognition is reflected in an intelligent search strategy that finds a solution without exhaustive brute-force search

Rationality

Rationality means that an **agent** acts in a sensible and purposeful way to achieve its goals. A rational agent will always use as much of its knowledge as possible to guide its behavior. Generally, there are two different views on rationality:

Optimal Rationality

An ideal agent, “given its belief-desire system, **optimizes** its choices.”

→ Not feasible in reality

Bounded Rationality

“Bounded rationality is rationality as exhibited by decision makers of **limited abilities**.”

Rationality depends on the **success criterion**, the agent’s **prior knowledge** of the environment, the **actions** the agent can perform, and **past percepts**.

MIT Encyclopedia of the Cognitive Sciences, Vernon (2014), Russell and Norvig (2010)

Emergent Systems

Emergence is “the rise of a system that **cannot be predicted** or explained from antecedent conditions” (Encyclopædia Britannica). The main goal of cognitive processing an emergent system is to maintain and extend its **autonomy**.

- Emergence is based on continuous **self-organization** of the system
- An emerging system develops its **own model** of the reality through **interaction** with the environment and **filtering** of relevant information
- In contrast to the computational functionalism of cognitivism, knowledge and representation are **specific** to the emerging system and depend on its history of interaction (→ **experience**)
- Knowledge is acquired in the form of additional **anticipatory skills**
- Emergent systems necessarily require **embodiment** and **development**

Connectionist Emergent Systems

Connectionist cognitive systems model cognition based on networks of **simple interconnected** computational units. Cognitive processing in these networks is **distributed, parallel**, and based on **statistical properties** instead of formal rules.

Main Components (Bechtel and Abrahamsen, 2002)

- Simple elements called **units**
- Equations that determine an **activation** value for each unit at each point in time
- **Weighted connections** between units which permit the activity of one unit to influence the activity of other units
- *Optionally:* **Learning rules** which change the network's behavior by changing the weights of its connections

Knowledge is stored in the connection weights of the system. The most prominent class of connectionist systems are **neural networks**.

Enactive Emergent Systems

An enactive emergent system “**develops** its **own understanding** of the world around it through its **interactions** with the environment” (Vernon, 2014).
Enaction is governed by two key mechanisms:

Phylogeny

The interaction between the system and its environment is **structurally determined** by the **innate** embodied physical and cognitive capabilities

Ontogeny

- The system is **structurally coupled** with the environment and **makes sense** of it based its **actions** the **perception** of resulting actions of the **environment**
- Through sense-making, the system generates its **own specific epistemology** that increases its cognitive capacity by capturing the **regularities** of interaction (→ **exploration**)
- The gradual increase of cognitive capacity results in **development**

Dynamical Emergent Systems

Dynamical emergent models of cognitive systems are based on **time-dependent differential equations** that allow for a **compact representation** of complex system behavior and the application of analysis methods from **dynamical systems theory** (**attractors**, **bifurcations**, **stability**, etc.).

Main Properties of Dynamical Systems

- **Dissipation**: the number of reachable states reduces over time
- **Non-Equilibrium System**: stable function requires an external “energy” supply
- **Non-Linearity**: complex behavior can emerge from a small set of state parameters
- **Collective Variables**: the system is represented by a small set of state variables

Dynamical Emergent Systems

Representation

Dynamical emergent systems can (but not necessarily do!) represent entities by means of **system states** or **trajectories**. These representations are **transient** and **context-dependent**.

The Role of the Factor Time

- In dynamical systems, time is an active factor of computation that co-determines the system behavior
- The **synchrony** between the agent and the environment imposes **constraints** on the **processing speed** of the system

Relation to Connectionist Systems

- Connectionist systems can be **dynamical systems** (e.g. through recurrent connectivity)
- Connectionist modeling addresses the **microscopic level**

Summary: Paradigms in Cognitive Science

Characteristic	Cognitivist	Emergent
Role of Cognition	Rational goal-achievement	Self-maintenance and self-development
Computational Operation	Syntactic manipulation of symbols	Concurrent self-organization of a network
Representational Framework	Patterns of symbols and tokens	Global system states
Temporal Constraints	Atemporal	Synchronous real-time entrainment
Embodiment	No role implied: functionalist	Direct constitutive role: non-functionalist
Perception	Abstract symbolic representations	Perturbation by the environment
Action	Causal result of a symbol manipulation	Perturbation by the system
Adaption	Learn new knowledge	Develop new dynamics
Motivation	Criteria for goal selection	Increase space of interaction

From Vernon (2014)